

GPU Speed Of Light Throughput

GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor.

|                             |       |                          |               |
|-----------------------------|-------|--------------------------|---------------|
| Compute (SM) Throughput [%] | 94.08 | Duration [ms]            | 46.20         |
| Memory Throughput [%]       | 94.08 | Elapsed Cycles [cycle]   | 71,371,879    |
| L1/TEX Cache Throughput [%] | 94.32 | SM Active Cycles [cycle] | 71,205,269.58 |
| L2 Cache Throughput [%]     | 12.78 | SM Frequency [Ghz]       | 1.54          |
| DRAM Throughput [%]         | 37.41 | DRAM Frequency [Ghz]     | 7.99          |

High Throughput

This workload is utilizing greater than 80.0% of the available compute or memory performance of this device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing workloads in the [Compute Workload Analysis](#) section.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

|                                        |           |                                              |                 |
|----------------------------------------|-----------|----------------------------------------------|-----------------|
| Grid Size                              | 4,096     | Function Cache Configuration                 | CachePreferNone |
| Registers Per Thread [register/thread] | 56        | Static Shared Memory Per Block [Kbyte/block] | 32.77           |
| Block Size                             | 256       | Dynamic Shared Memory Per Block [byte/block] | 0               |
| Threads [thread]                       | 1,048,576 | Driver Shared Memory Per Block [Kbyte/block] | 1.02            |
| Waves Per SM                           | 56.89     | Shared Memory Configuration Size [Kbyte]     | 102.40          |
| Uses Green Context                     | 0         | Stack Size                                   | 1,024           |
| # SMs [SM]                             | 24        | # TPCs                                       | 12              |
| Enabled TPC IDs                        | all       | -                                            | -               |

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

|                                        |       |                                |    |
|----------------------------------------|-------|--------------------------------|----|
| Theoretical Occupancy [%]              | 50    | Block Limit Registers [block]  | 4  |
| Theoretical Active Warps per SM [warp] | 24    | Block Limit Shared Mem [block] | 3  |
| Achieved Occupancy [%]                 | 49.75 | Block Limit Warps [block]      | 6  |
| Achieved Active Warps Per SM [warp]    | 23.88 | Block Limit SM [block]         | 24 |

Theoretical Occupancy

Est. Local Speedup: 50.00%

The 6.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (50.0%) is limited by the required amount of shared memory.

Key Performance Indicators

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

|                                    |               |                                    |               |
|------------------------------------|---------------|------------------------------------|---------------|
| Average SM Active Cycles [cycle]   | 71,205,269.58 | Average L1 Active Cycles [cycle]   | 71,205,269.58 |
| Average L2 Active Cycles [cycle]   | 66,276,489.31 | Average SMSP Active Cycles [cycle] | 71,205,245.07 |
| Average DRAM Active Cycles [cycle] | 138,140,456   | Total SM Elapsed Cycles [cycle]    | 1,713,131,920 |
| Total L1 Elapsed Cycles [cycle]    | 1,713,131,920 | Total L2 Elapsed Cycles [cycle]    | 1,064,336,880 |
| Total SMSP Elapsed Cycles [cycle]  | 6,852,527,680 | Total DRAM Elapsed Cycles [cycle]  | 1,476,960,256 |

Missing [Roofline](#) and [Memory Charts](#)? Profile again with the *detailed* [metric set](#) to collect all necessary metrics. Also consider collecting all available sections with the *full* metric set.